

# Project Proposal

## **Name of the Project:**

Multi-Agent AI Framework for Task-Aware Model Routing and Execution

## **The URL of the Project Web Page:**

Project documentation website url : <https://multi-agent-ai-framework.mintlify.app/>

Project website url : <https://multi-agent-ai-framework.vercel.app/>

## **Names of the Team Members:**

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## **Project Description :**

This project focuses on the design and implementation of a configurable multi-agent AI assistant framework that orchestrates multiple specialized AI components through a unified conversational interface. Instead of operating as a single-purpose chatbot, the system is structured as a modular framework in which different agents are responsible for distinct functional domains such as programming assistance, repository inspection, technical search, workflow automation, and speech-based interaction. A central coordination component interprets user requests and determines how these agents should be combined to produce a coherent response.

The platform is designed to support both on-premises and cloud-based AI models, allowing different execution backends to be selected depending on performance, cost, and availability constraints. On-premises deployment refers to running models on users own server hardware within their local network, operating independently without requiring external internet connectivity, ideal for organizations with strict data privacy requirements or air-gapped environments. This hybrid approach makes the system suitable for both lightweight local usage and more demanding tasks that require external cloud services.

In addition to these options, users can connect their own trained or fine-tuned models by providing OpenAI-compatible endpoints. This allows researchers, developers, and organizations to deploy models on their own infrastructure whether on-premises servers, private cloud instances, or dedicated GPU machines maintaining complete ownership and control over their AI capabilities.

By separating coordination logic, agent logic, and execution backends, the project aims to provide a flexible and extensible foundation for building intelligent, task-oriented AI applications.